B.Tech (ME)– III-II Sem

23A03602P	CAD/CAM Lab (Professional Core)	L	T	P	C
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Course objectives: The objectives of the course are to	
1	Develop students' skills in drafting and understanding orthographic, isometric views, and CAD file formats like DXE and IGES.
2	Enable the creation of 3D part models using basic and advanced features in CAD tools.
3	Hands-on experience in assembly modeling using both feature-based and Boolean-based methods.
4	Familiarize with CAM software for generating NC code for various machining processes.
5	Expertise industrial manufacturing via the use of post-processors and NC machines for real-time machining.

Course Outcomes: On successful completion of the course, the student will be able to,		
1	Create accurate 2D technical drawings using orthographic and isometric projections, and interpret file formats like DXE and IGES.	*L1,L2,L3*
2	Develop 3D part models using features such as protrude, revolve, shell, and sweep, and demonstrate and evaluate parent-child relationships.	*L2,L3,L5*
3	Construct and analyze assemblies using feature-based and Boolean operations for simple mechanical systems and create complex mechanical assemblies.	*L3,L4,L6*
4	Generate NC code for complex surfaces using CAM tools and study the function of various post-processors.	*L1,L3,L6*
5	Transfer NC code to CNC machines. Perform and evaluate basic machining tasks such as turning and milling and generate new technics by using modern tools.	*L4,L5,L6*

List of Experiments:

1. **Drafting:** Development of part drawings for various components in the form of orthographic and isometric. Representation of Dimensioning and tolerances scanning and plotting. Study of script, DXE AND IGES FILES.
2. **Part Modelling:** Generation of various 3D Models through Protrusion, revolve, shell sweep. Creation of various features. Study of parent child relation.
3. **Assembly modelling:** Feature based and Boolean based modelling surfaces, Assembly Modelling of simple components and Design of simple components.
4. CAM**:**

a) Study of various post processors used in NC Machines.
b) Development of NC code for free form and sculptured surfaces using CAM packages.
c) Machining of simple components on NC lathe and Mill by transferring NC Code / from a CAM packages.

Through Any Four Software Packages from the following: Use of Auto CAD, Micro Station, CATIA, Pro-E, I-DEAS, , CAEFEM, Gibbs CAM, Master CAM etc,.

5. Evaluation of Stress/Strain for a plate with a hole.